
EMPLOYMENT

Lab Instructor	University of Toledo	August 2014 – May 2016
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- Supervising design and implementation of buffering, de-multiplexing, decoding and memory subsystems (RAM & ROM) circuitry for 80186XL microprocessors.
- Supervising software projects (development, testing and debugging of assembly code) on designed micro-controller system (embedded system).

Consultant	Mon-Say Corp (Tech Startup)	May 2014 – August 2014
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- Designed & implemented (using embedded C on Keil µVision IDE) a fall detection machine learning algorithm (Random Forest) on iNEMO-Inertial Modules (ARM Cortex M3 based microcontroller with accelerometer & gyroscope sensor).

Specialist, Automation	Robi Axiata Ltd	April 2009 – July 2013
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- Developed Web based tools (using **JAVA**/JSP, Oracle) to support Network surveillance and O&M.
- Developed PL/SQL stored procedures in Oracle database environment.
- Developed data loader (move data between MySQL & Oracle database) windows applications using **JAVA**.
- Developed MML (Man Machine Language) automation tools using **JAVA** socket programming.

EDUCATION

Toledo, OH	University of Toledo	Fall 2013 – Spring 2016
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- M.S.C. in Electrical Engineering. GPA: 3.86 (Up to 4th Semester)
- Graduate Coursework: Artificial Intelligence; Computer Architecture; Computer Vision & Image Processing; Bio-Inspired Computing; Human Computer Interface; Statistical Computing.

Khulna, Bangladesh	Khulna University of Engineering & Technology	April 2002 – March 2006
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- B.S.C. in Electrical & Electronics Engineering. GPA: 3.02.
- Undergraduate Coursework: Microprocessor and Microcomputer; Advanced Microprocessor & Microcomputer; Programming Technique I & II; Electrical Circuits I, II & III; Electrical Machines I, II & III; Electronics I, II & III; Digital Electronics; Communication Engineering I, II & III.

TECHNICAL EXPERIENCE

Projects

- **Interface project with speech recognition** (spring 2014). Designed (both MVC & PAC system architectures) and implemented a calculator with multi modal interface (Speech, Mouse Click and Keyboard) for input and output as a part of one of the graduate course work project. Implemented the code in **JAVA** as well used CMU Sphinx as JAVA speech API.
- **Real time simulation of planning and scheduling algorithm of multi-storied automated robotic parking system** (Master's thesis project). For my master's thesis, I had to conduct research to obtain efficient and robust real time elevator scheduling algorithm feasible for specific multistoried automated robotic parking systems. In order to verify and test the efficiency of real time algorithms in automated multi storied robotic parking lot environment, I had to develop certain simulation model for real time and distributed systems. I used "multi-threading programming" in **JAVA** in order to developed simulation software (wrote around **8000 lines of JAVA code** using **eclipse IDE**). I also needed to work with "Semaphore" techniques for proper synchronization and memory resources sharing among thousands of real time parallel threads.

LANGUAGES AND TECHNOLOGIES

- C++; C; Java; Excel VBA; PL/SQL; JavaScript; HTML; JSP
- Visual Studio; Oracle SQL Server; Eclipse; Keil µVision; ARM Cortex M3 C Libraries, ORCAD Capture, ORCAD Layout